FIELD EYES v2 – solar soil–monitoring node with latching valve control

POWER TREE
6 V panel (1–2 W) –> CN3791 MPPT charger –> 18650 cell (DW01A/FS8205 protection) = VSYS 3.0–4.2 V
VSYS –> A7672S VBAT directly (2 A bursts)   VSYS –> XC6220 –> +3V3 (ESP32, always on)
+3V3 –> AO3401A (SENS_EN) –> +3V3_SW (RS485, DHT22, NTC)
VSYS –> AO3401A (SENS_EN) –> MT3608 –> +12V_SENS (soil sensors) –> 78L05 –> +5V_SENS (MAX485) and –> 78L09 –> 8.7 V valve reservoir (MX1508)

WHAT CHANGED FROM THE OLD BOARD (security.sch)
- MP2338 buck, LM1117 and the D6/D7 diode drop removed: the modem runs straight from the cell.
- Solar MPPT charging and cell protection added; battery voltage, temperature and charge state readable.
- PWRKEY no longer tied to GND (the modem can stay off); RI wired for wake–on–message.
- ESP32 TX –> modem RX connection fixed (Q2 collector was not connected).
- RS485 keeps the MAX485ESA+T (5 V from a 78L05 on the sensor rail); sensors and RS485 switched off between readings.
- The valve driver chip (MX1508, from the mini–L298N module) is on the board; screw terminals for field cables.
- EN reset delay and auto–reset programming added; GPS antenna kept (U.FL J304, active–antenna bias as on the old board).

PCB LAYOUT NOTES
- Solid ground plane on the bottom layer; keep the CN3791 loop (Q102, D101, D102, L101, C101) and the MT3608 loop tight.
- VSYS –> FB301 –> modem VBAT: >= 2 mm wide, C301–C306 at the VBAT pins. Cell holder close to the modem.
- RF_ANT –> J301: 50 ohm coplanar trace, as short as possible, ground vias along both sides.
- Keep the ESP32 antenna end over a board edge with no copper underneath (or cut–out).
- TH101 under the 18650; SM712s right at the screw terminals; C403 and U403 next to the valve terminal.

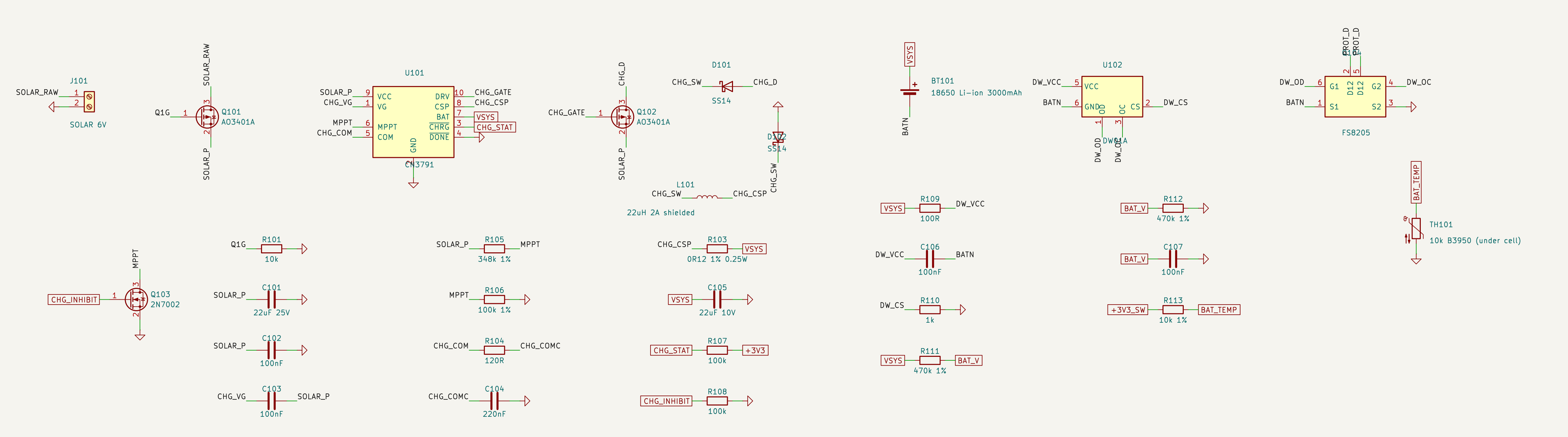
1 POWER: SOLAR CHARGER, BATTERY, 3.3 V AND SWITCHED RAILS

A. SOLAR INPUT + MPPT CHARGER (CN3791, 1 cell, 4.2 V)

Panel: 6 V nominal, 1–2 W (Voc <= 8 V). Q101 = reverse–polarity protection (on when the panel is the right way round).
Charge current = 120 mV / R103 = 1.0 A max (the panel limits it further). MPPT holds the panel at
1.205 V x (1 + R105/R106) = 5.40 V, a little under Vmp so it still tracks when the panel is hot.
CHG_INHIBIT high (Q103 on) pulls MPPT low and stops charging – firmware uses it above 45 C or below 0 C
(the CN3791 has no temperature input). /DONE unused –> GND (datasheet).

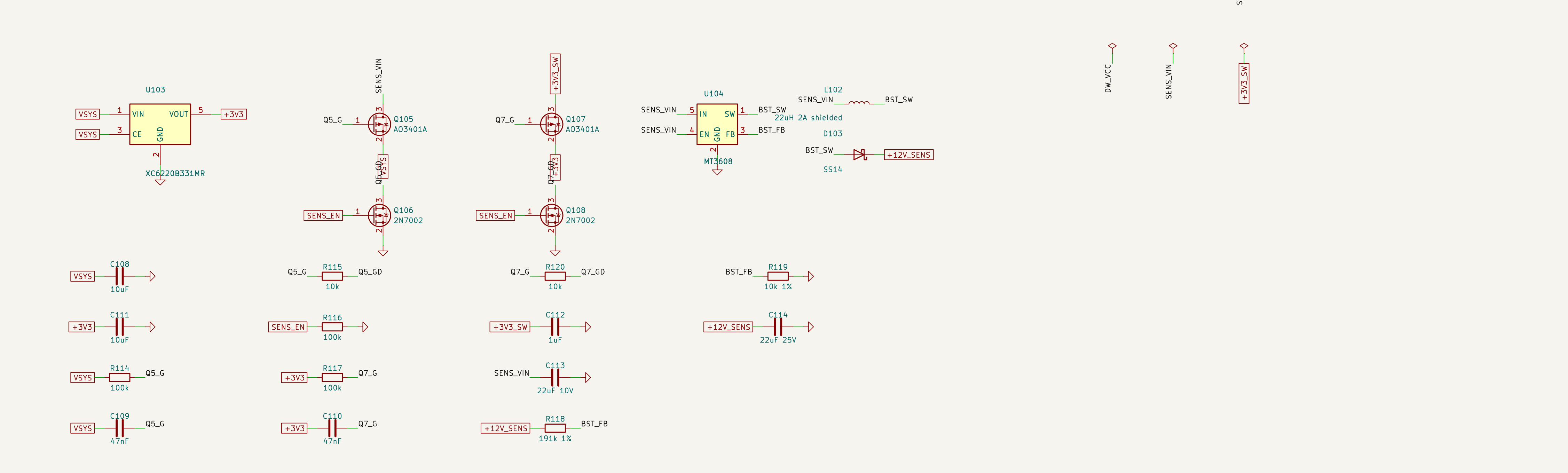
B. BATTERY (1 x 18650 Li–ion) + PROTECTION + SENSING

DW01A + FS8205 cut the cell off below 2.4 V, above 4.3 V, on over–current (~3 A) and short circuit.
Board ground is the protected P– side; BATN is the cell minus. VSYS = cell + (3.0–4.2 V) feeds the
modem directly (SIMCom reference design: no regulator, no diodes), the 3.3 V LDO and the 12 V boost.
BAT_V = VSYS / 2 (470k/470k, 4.5 uA). TH101 sits under the cell; its divider is powered from +3V3_SW
so it draws nothing while the sensors are off.



C. 3.3 V REGULATOR, SWITCHED SENSOR POWER (AO3401A) AND 12 V SENSOR / VALVE BOOST

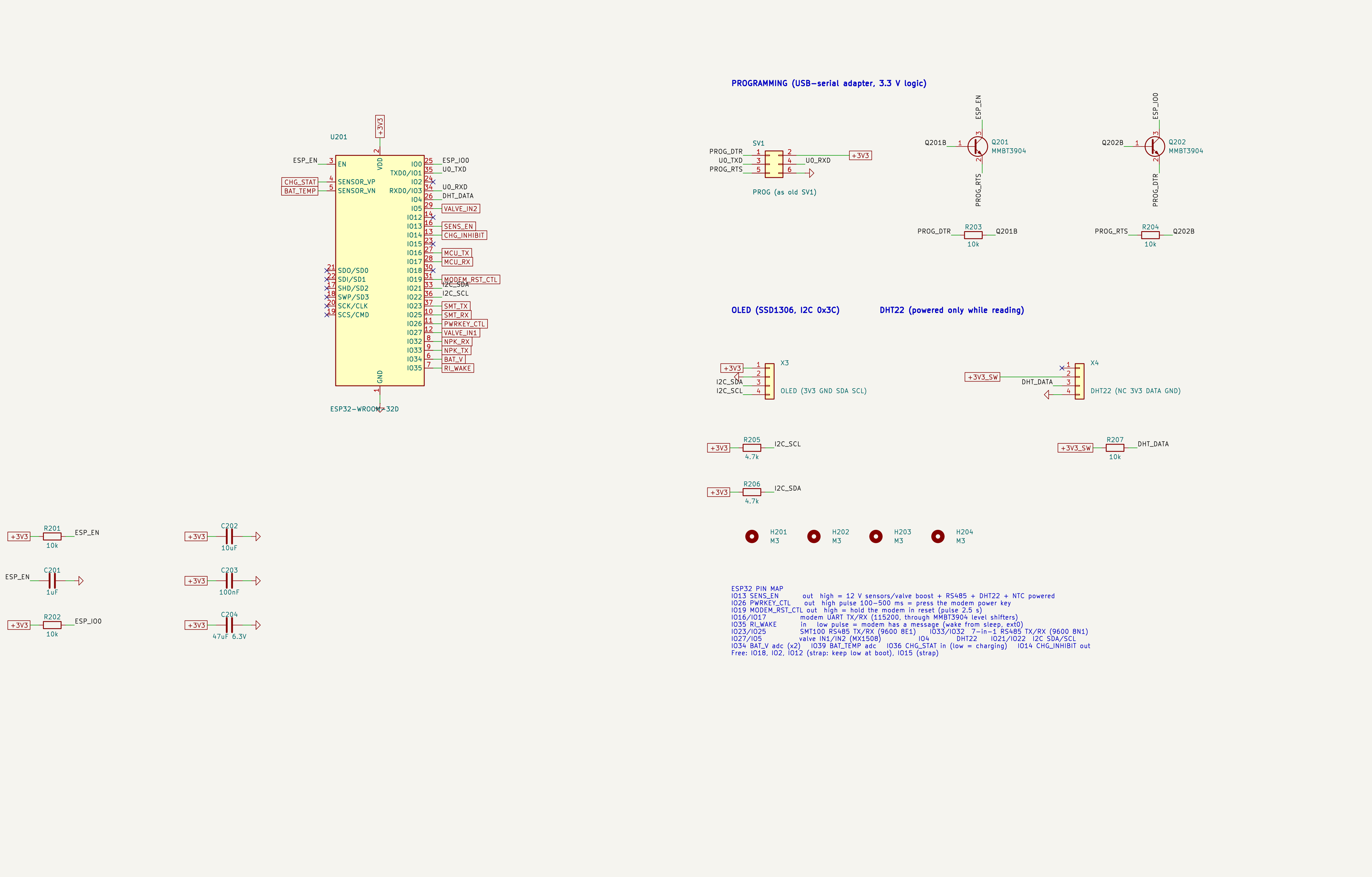
U103 XC6220: 1 A, 8 uA quiescent, ~0.1 V dropout –> the ESP32 still gets ~3.2 V from a 3.3 V cell.
SENS_EN (GPIO13) high turns on BOTH Q105 (cell –> boost) and Q107 (+3V3 –> RS485/DHT22/NTC). Off by default:
R116 holds SENS_EN low while the ESP32 boots or sleeps. Each AO3401A gate is pulled to its own source
(100k) and pulled down through 10k by a 2N7002, because an ESP32 pin (3.3 V) cannot switch a P–FET
whose source sits at 4.2 V. C109/C110 slow the turn–on (~0.5 ms) so the inrush cannot trip the DW01A.
The boost input is switched (not its EN pin): an MT3608 that is merely disabled still passes the cell
voltage to its output through the inductor and diode, and draws ~100 uA. Vout = 0.6 x (1 + R118/R119) = 12.06 V.



2 ESP32, PROGRAMMING, DISPLAY, DHT22

ESP32–WROOM–32D

EN: 10k + 1 uF delay (Espressif reference) so the chip starts only after 3.3 V is stable.
Q201/Q202: the usual two–transistor auto–reset, so esptool can reset and enter the bootloader through
DTR/RTS of a USB–serial adapter (same pin order as the old SV1 header). Strapping pins IO0/IO2/IO12/IO15
are left free or pulled the safe way. Flash pins (SD0–SD3, CLK, CMD) must stay unconnected.
Do not connect the adapter's 3.3 V while the battery is fitted (two regulators would fight).
There are no reset or boot buttons: reset and boot mode come from the programming header SV1 (RTS –> EN, DTR –> IO0).



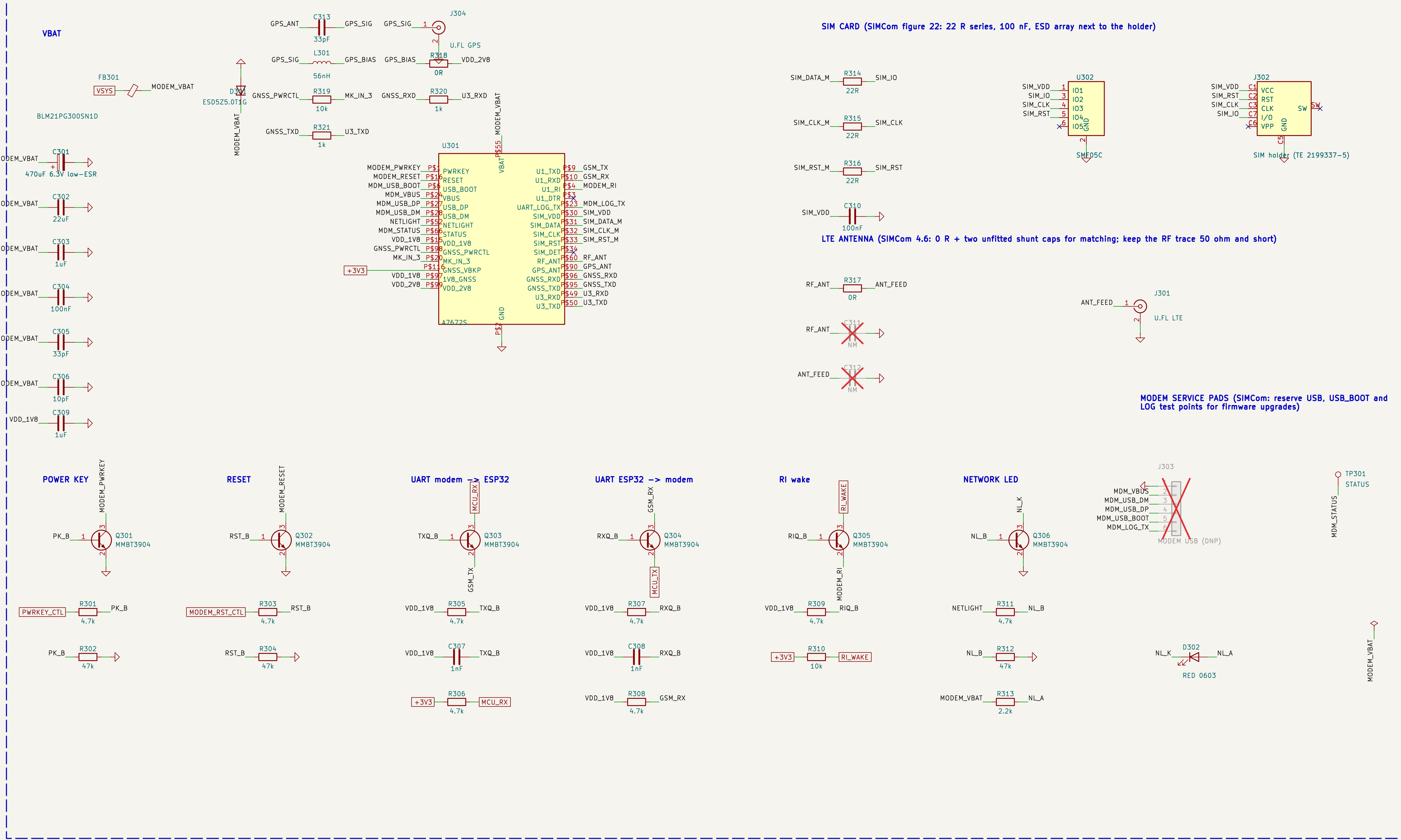
ESP32 PIN MAP
IO13 SENS_EN   out high = 12 V sensors/valve boost + RS485 + DHT22 + NTC powered
IO26 PWRKEY_CTL   out high pulse 100–500 ms = press the modem power key
IO19 MODEM_RST_CTL out high = hold the modem in reset (pulse 2.5 s)
IO16/IO17   modem UART TX/RX (115200, through MMBT3904 level shifters)
IO35 RI_WAKE   in   low pulse = modem has a message (wake from sleep, ext0)
IO23/IO25   SMT100 RS485 TX/RX (9600 8E1)   IO33/IO32 7–in–1 RS485 TX/RX (9600 8N1)
IO27/IO5   valve IN1/IN2 (MX1508)   IO4   DHT22   IO21/IO22 I2C SDA/SCL
IO34 BAT_V adc (x2)   IO39 BAT_TEMP adc   IO36 CHG_STAT in (low = charging)   IO14 CHG_INHIBIT out
Free: IO18, IO2, IO12 (strap: keep low at boot), IO15 (strap)

3 A7672S LTE MODEM, SIM, ANTENNA

A7672S LTE CAT–1 MODEM

Powered straight from the cell (VSYS) through FB301: SIMCom reference for a 1–cell Li–ion supply. 470 uF low–ESR
plus 22u/1u/100n/33p/10p near the VBAT pins (route VBAT >= 2 mm wide). D301 clamps surges.
PWRKEY and RESET are driven by MMBT3904s (SIMCom figures 10 and 13: 4.7k base, 47k pull–down), so the
modem stays OFF until the firmware presses PWRKEY. UART level shifters: SIMCom figure 16 (MMBT3904, 4.7k, 1 nF).
RI goes to an RTC pin of the ESP32 so it can sleep and wake on every incoming MQTT message.
Audio, SPI, camera and keypad pins are not used and stay unconnected (not drawn on the symbol). GPS: see the GPS ANTENNA block.

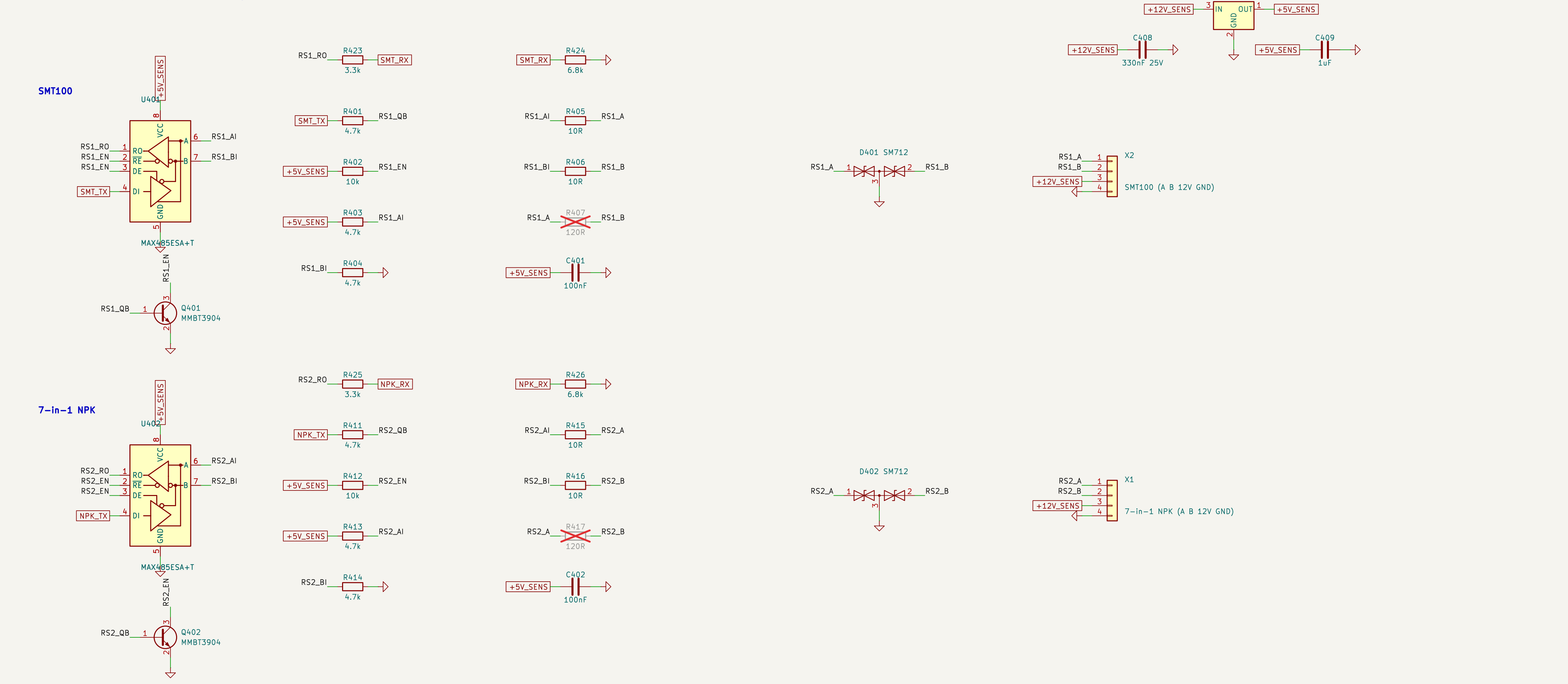
GPS ANTENNA (active antenna, as on the old board): the modem's 2.8 V feeds the antenna through R318 + L301 (56 nH); C313 blocks the DC.
The built–in GNSS UART is linked to UART3 through R320/R321 and GNSS_PWRCTL goes to MK_IN_3 through R319 (SIMCom 3.11). 1V8_GNSS and GNSS_VBKP are powered.
For a passive antenna, fit C313 only and do not fit R318/L301 (and send AT+CVAUXS=0).



4 SOIL SENSORS (RS485) AND VALVE DRIVER

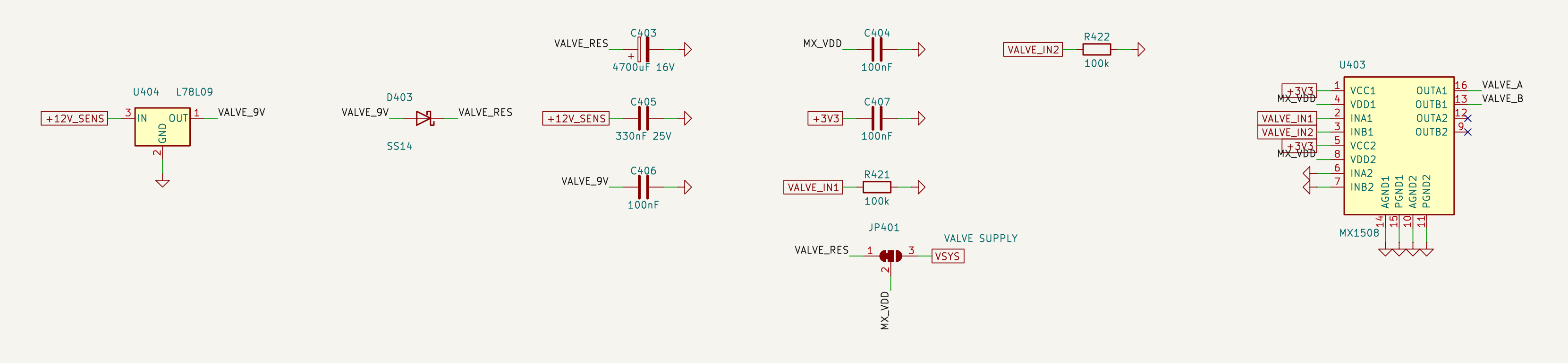
RS485 SOIL SENSORS (two buses: the SMT100 is 8E1, the 7–in–1 8N1)

MAX485ESA+T (the chip on the old board) on +5V_SENS, made by U405 (78L05) from the 12 V sensor rail, so it only has power
while SENS_EN is on. RO swings 0–5 V: the 3.3k + 6.8k divider gives the ESP32 3.4 V (its pins are not 5 V tolerant).
Direction is automatic: TX idle–high turns the MMBT3904 on and holds DE/RE low (receive); a 0 bit releases DE so the driver
pulls the bus low; 1 bits come from the 4.7k bias. 10 R + SM712 protect the field cable. 120 R not fitted (short cable, 9600 baud).
Connectors X1 (7–in–1) and X2 (SMT100) are 4–pin JST XH, pin 1 A, 2 B, 3 +12 V, 4 GND, like the old board.
Firmware: set the TX pins low/input before switching the sensors off, or they feed the unpowered chips.



LATCHING VALVE DRIVER (MX1508 – the chip on the "mini L298N" module)

U403 = MX1508, channel 1: IN1 (GPIO27) high = OUTA high (open); IN2 (GPIO5) high = OUTB high (close); both low = standby
< 0.1 uA. Same logic as the module, so the firmware does not change. Its motor supply must stay below 10 V, so the 12 V boost
charges C403 through a 78L09 (current–limited, ~0.1 A) and D403 to ~8.7 V; C403 then supplies the coil pulse, and D403 keeps it
charged while the boost is off. JP401 (solder jumper): 1–2 = 8.7 V reservoir (9 V valves, default), 2–3 = cell (3.6–6 V valves).
Firmware: SENS_EN on >= 1 s before a pulse (tops up C403). Logic supply = +3V3, so the 3.3 V inputs are always in range.



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Field Eyes
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